www.csvtuonline.com

Roll No. : .....

324452(24)

B. E. (Fourth Semester) Examination,
April-May, 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTRO MAGNETIC THEORY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) from each question is compulsory. Solve any two parts from (b), (c) and (d).*

(a) Write two application of cross product. 2

(b) A closed surface is defined in cylindrical coordinate by $4 \le \rho \le 6$, $30° \le \phi \le 60°$, $2 \le z \le 5$

Find :

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(i) The volume enclosed.

(ii) What is the length of longest straight line th lies entirely within the volume?

(iii) What is the total area of the surface?

(iv) Find the total length of twelve edges of tl surfaces.

(c) Find the expression for electric field intensity due t infinite line charge distribution available in the z-a

(d) Find the electric field at origin if the following charg distribution are present in the space. Point charg 12 nC at $P(2, 0, 6)$. Uniform line charge densit 3 nC/m at $x = -2$, $y = 3$. Uniform surface charg density 0·2 nC/m^2 at $x = 2$.

2. (a) What do you mean by divergence of a vector fiel

(b) If $\overline{G} = 5 r \sin^2 \theta \cos^2 \phi \, \overline{a_r}$ evaluate both side divergence theorem for the region $r \le 2$

(c) Write the application of Gauss's law for :

(i) Uniform line charge distribution

(ii) Uniform surface charge distribution

www.csvtuonline.com

www.csvtuonline.com

(d) In a field

$$\overline{E} = \left(-50\, y\, \overline{a_x} - 50\, x\, \overline{a_y} + 30\, \overline{a_z}\right) \text{V/m}$$

Calculate the differential amount of work done in moving a $2\,\mu C$ charge over a distance of $5\,\mu m$ from $A(1,2,3)$ to $B(2,4,1)$.

3. (a) Explain uniqness theorem.

(b) Two perfect dielectric have $\varepsilon r_1 = 2$ and $\varepsilon r_2 = 8$ the planer interface between them is the surface $x - y + 2z = 5$. The origin lies in the region 1 if

$\overline{E_1} = 100\,\overline{a_x} + 200\,\overline{a_y} - 50\,\overline{a_z}$ V/m find $\overline{E_2}$.

(c) Given the potential field $V = \dfrac{50 \sin\theta}{r^2}$ volt in free space.

(i) Determine whether V satisfied Laplace equatio

(ii) Find the total charge stored inside the spherical shell $1 \le r \le 2$

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(d) Derive the relation between $\overline{J}$ and ρv state and explain continuity equation of current in integral form and point form. 7

4. (a) What is Lorentz force equation? 2

(b) In a cylindrical coordinate, a magnetic field is given as $\overline{H} = \left(2\rho - \rho^2\right)\overline{a_\phi}$ A/m, $0 \le \rho \le 1$.

(i) Determine the current density as a function of ρ within this cylinder

(ii) What total current pases through the surface $\rho = 0,\ 0 \le \rho \le 1$

(iv) Check (ii) by another method. 7

(c) Derive magnetic boundary condition between two magnetic media. Also derive expression for refraction law for magnetic flux. 7

(d) Find the magnetic field intensity within a magnetic material where

(i) $M = 150$ A/m and $\mu = 1{\cdot}5 \times 10^{-5}$ H/m

(ii) $\overline{B} = 300\,\mu T$ and $\chi_m = 15$

www.csvtuonline.com

www.csvtuonline.com

(iii) there are $8{\cdot}2 \times 10^{28}$ atom/m^3, each atom has a dipole moment of $5 \times 10^{-27}\ A-m^2$ and $\mu r = 30$. 7

5. (a) Explain Faraday's law. 2

(b) Assume a homogenious material of infinite extent with $\varepsilon = 2 \times 10^{-10}$ F/m, $\mu = 1{\cdot}25 \times 10^{-5}$ H/m and $\delta = 0$. Let $\overline{E} = 400 \cos\left(10^9 t - kz\right)\overline{a_x}$ V/m if all the fields vary sinusoidally (or cosinusoidally) use Maxwell's equation to find $\overline{D}, \overline{B}, \overline{H}$ and K. 7

(c) A circular loop conductor lies on $z = 0$ plane and has a radius of 0·1 m and resistance of $\frac{5\,\Omega}{a_z}$. Given $\overline{B} = 0{\cdot}2 \sin 10^3 t\, \overline{a_z}$ T. Determine the current in the loop. 7

(d) Explain displacement current density. Derive Maxwell equation related to displacement current density in differ initial and Integral from. 7

www.csvtuonline.com

www.csvtuonline.com